

Towards Enhancing the Effectiveness of Osun Health Insurance Scheme: A Mixed Method Study of Human Resource Capacity of Healthcare Facilities in Osun State

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ABSTRACT

Health insurance is an avenue for achieving universal health coverage and key sustainable development goals. Osun Health Insurance Scheme (OHIS) was established to bridge the gap in health inequalities and health financing. This study aimed to assess the level of manpower in the healthcare facilities with and without OHIS accreditation in Osun State. This cross-sectional descriptive mixed method study, conducted between January to March 2022, incorporated private and government facilities. Quantitative data was collected with a semi-structured questionnaire, while qualitative data was collected with key informant interviews from relevant health professionals. Statistical analysis was done using Stata version 16, and p value was significant at <0.005. Majority of health facilities were private (82, 75.2%), and there was no statistically significant difference in the organization of the healthcare services between OHIS accredited and unaccredited facilities. Accredited health facilities, private or government, had more permanent staff while unaccredited facilities had more temporary staff; more accredited government facilities had more permanent staff (20 or more) than unaccredited facilities (61.5% vs 15.4%, $p=0.016$), while more unaccredited government facilities had more temporary staff than accredited facilities (76.9% vs 15.4%, $p=0.047$). OHIS provided extra fund for staffing and consumables. Qualitative data revealed shortage of staff in the government facilities, and that OHIS has strengthened workforce with impact on care. There is shortage of health professionals in Osun State with possibilities of improvement with OHIS. The OHIS should strive to include more healthcare facilities, for improved quality of care of the populace.

Keywords: OHIS, Osun State, Healthcare Facilities, Accredited, Staff

INTRODUCTION

Access to healthcare is a big challenge in Nigeria and despite some improvement in health over the past two decades, Nigeria still has poor key health indicators; high rates of maternal and infant mortality and under-five mortality and low life expectancy are realities of the country's poor healthcare system.¹ The key challenges with healthcare utilization in Nigeria are poor health financing and prohibitive cost of healthcare felt by the vast majority of citizens.² This health burden can be mitigated to great extent by leveraging the

health insurance system to enhance access to healthcare across all strata in the society. Health Insurance Schemes provide a pathway to achieving Universal Health Coverage (UHC).³

The Nigeria National Health Insurance Scheme (NHIS) was established for health need of the citizenry, it has barely covered less than 8% of the 208 million population of the country achieved even after 20 years of operation.⁴ The increasing need for subnational health insurance in Nigeria was because of the over-centralized nature of the NHIS with little or no structure in place to meet the health needs of the teaming population at the grass-root. Osun Health Insurance Scheme (OHIS) is an avenue for subnational engagement of populace within the catchment. The OHIS offers a comprehensive health plan benefit package that addresses health care priorities related to the prevalent health conditions in

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the state. The Scheme provides primary, secondary and tertiary care services in about 120 accredited facilities across all the local government areas in the state.⁵ Unlike unaccredited facilities, accredited facilities had undergone initial pre-enlistment audits and substantially met set criteria for inclusion to the insurance scheme. They have demonstrated adherence to minimum set standard of care quality and patient safety. Accreditation process is periodic and involves ongoing monitoring for continuous quality improvement. Accredited facilities enjoy more patients' patronage not only because of the subsidized service, but also due to people's confidence in their capacity to provide good quality care.

For provision of sustained qualitative and accessible healthcare services, the coordinating agencies for social health insurance must mandate adequacy of capacities, especially in the area of manpower, for functional healthcare facilities within the context of the current realities of healthcare demands. Healthcare workers are a critical component of health system and crucial to achieving the health-related targets of Sustainable Development Goals (SDGs). There is a projection of a shortfall of 10 million healthcare workers by 2030 with greater proportion and impact in low-income countries.⁶ In addition, Nigeria has particularly suffered decline in medical manpower in recent years due to mass exodus for greener pastures.^{7,8} This study therefore assessed the adequacy of human resources according to facility ownership and accreditation status.

MATERIALS AND METHODS

Study Area

This study was conducted in Osun State, located in South-West Nigeria, with a projected population of 5,134,434, who are predominantly farmers, traders and civil servants. There are 1095 total registered healthcare facilities spread across the 30 local government areas (LGAs).

Study Design

This study was a cross-sectional descriptive study which included OHIS

accredited and unaccredited or yet-to-be accredited facilities, and carried out between January and March 2022. It utilized a mixed-method of quantitative and qualitative data collection approaches.

Study Location and Selection of Study Facilities

The study was carried out in 109 OHIS accredited and unaccredited private and government health facilities; selected from a total of 1,037 in the 28 LGAs distributed as 723 private and 314 government facilities. Two LGAs were excluded because of sparse accredited facilities. Because private health facilities were more in number compared to government facilities, and more unaccredited private facilities compared to accredited private facilities, the selection of accredited to unaccredited private facilities was in a ratio of 1:1.3 and accredited and unaccredited government facilities in a ratio of 1:1. Thus, a total of 109 healthcare facilities recruited were 83 private facilities (including 36 accredited and 47 unaccredited facilities); and 26 government facilities (including 13 accredited and 13 unaccredited facilities). The facilities were selected by purposive sampling.

Data Collection

Quantitative data were collected from the heads of the accredited and unaccredited health facilities or the most senior medical personnels in the facility. The data were collected by with a purpose-designed semi-structured questionnaire.

Qualitative data was collected with key informant interview (KII) guide among key officials and personnel of accredited health facilities. The interview, lasting for about 30 minutes, was designed around five themes: adequacy of manpower, effects and consequences of manpower shortage, impact of OHIS on staffing, impact of OHIS on quality of care, and influence on OHIS on availability of consumables and supplies.

Data Analysis

Data analysis was done using Stata version 16 (Stata Corp., Texas). Findings were presented using frequency tables. Inferential

statistics was done with chi square and p value was significant at <0.05 .

All recorded sessions were transcribed verbatim and interviews conducted in the local dialects were translated into English Language. The transcripts from the KIIs were analyzed, with a focus on identifying recurrent, dominant, and divergent opinions. The findings were organized around themes and sub-themes with the ATLAS. qualitative analysis software version 8.

Ethical Considerations

Ethical clearance was obtained from the Ethics Committee of the Institute of Public Health, Obafemi Awolowo University (HREC Number: IPH/OAU/12/1786). Written informed consent and verbal consent were obtained from all participants and heads of the facilities respectively.

RESULTS

Sample Characteristics and Organization of Services in the Selected Health Facilities

The characteristics of the facilities selected for this assessment are presented in Table 1. About half, 26 (53.1%) of the 49 accredited facilities and 32 (53.3%) of the 60 unaccredited facilities were selected from the

Osun Central Senatorial District. Majority (82, 75.2%) of the hospitals were private comprising 36 (43.9%, 36/82) accredited and 46 (56.1%, 46/82) unaccredited facilities. Almost all (96.0%, 24/25) of the public hospitals were owned by the state government. There was no statistically significant difference in the distribution between accredited and unaccredited facilities across the three senatorial districts in the state.

The organization of health services in the selected facilities is shown in Table 2. Almost all, 45 (91.8%) of the accredited and 48 (80.0%) of the unaccredited facilities provided both in-patient and outpatient care; 44 (89.8%) of the accredited and 48 (80.0%) of the unaccredited operated every day of the week. In addition, forty-two (85.7%) of the accredited and 44 (73.3%) of the unaccredited opened for 24 hours. Of the facilities providing in-patient care, 25 (55.6%) accredited and 36 (72.0%) unaccredited had less than 20 beds while 39 (86.6%) of such accredited facilities and 48 (96.0%) of the unaccredited facilities dedicated only one to nine of their in-patient beds to maternity care. There was no statistically significant difference in the ways the healthcare services were organized between accredited and unaccredited facilities in the state.

Table 1: Senatorial districts, types and ownership of the selected healthcare facilities

	Accreditation status			Statistics
	Accredited (n = 49) n (%)	Unaccredited (n=60) n (%)	Total (n = 109) n (%)	
Senatorial districts:				$\chi^2=0.001$ df=2
Central	26 (53.1)	32 (53.3)	58 (53.2)	p value= 0.980
East	11 (22.4)	14 (23.3)	25 (22.9)	
West	12 (24.5)	14 (23.3)	26 (23.9)	
Facility type:				$\chi^2=0.026$ df=1
Private	36 (73.5)	46 (76.7)	82 (75.2)	p value= 0.871
Public	13 (26.5)	14 (23.3)	27 (24.8)	
Ownership:				$\chi^2=0.135$ df=1
State Government	13 (26.5)	13 (21.7)	26 (23.9)	p value= 0.553
Private	36 (73.5)	47 (78.3)	83 (76.1)	

Table 2: Organization of services in the selected facilities

Variables	Accreditation Status		Total (n=109) n (%)	Statistics
	Accredited(n=49) n (%)	Unaccredited (n=60) n (%)		
Type of care				$\chi^2=3.017$
In and outpatient	45 (91.8)	48 (80.0)	93 (85.3)	df=2
Inpatient only	0 (0.0)	2 (3.0)	2 (1.8)	p value=0.082
Outpatient only	4 (8.2)	10 (17.0)	14 (12.8)	
Number of days facility opens in a week				$\chi^2=1.966$
7 (every day)	44 (89.8)	48 (80.0)	92 (84.4)	df=1
Less than 7	5 (10.2)	12 (20.0)	17 (15.6)	pvalue= 0.163
Operating hours				$\chi^2=2.484$
24 hours	42 (85.7)	44 (73.3)	86 (78.9)	df=1
Less than 24 hours	7 (14.3)	16 (26.7)	23 (21.1)	p value=0.115
*Operating hours if less than 24 hours (n = 23)				$\chi^2=0.289$
4 – 8 hours	4 (57.1)	11 (68.8)	15 (65.2)	df=1
8 – 12 hours	3 (42.9)	5 (31.2)	8 (34.8)	p value=0.591
*Number of days facility is opened if less than 7 (n=18)				
5 days	5 (100.0)	12 (100.0)	17 (100.0)	
*For facilities offering in-patient care, the number of overnight beds (n = 95)				$\chi^2=1.739$
Less than 10	11 (24.4)	10 (20.0)	21 (21.1)	df=2
10 – 19	14 (31.1)	26 (52.0)	40 (42.1)	p value=0.841
20 or more	20 (44.4)	14 (28.0)	34 (35.8)	
*Number of beds dedicated to maternity care (n = 95)				$\chi^2=2.675$
1 – 9	39 (86.6)	48 (96.0)	87 (91.6)	df=1
10 or more	6 (13.3)	2 (4.0)	8 (8.4)	p value=0.143

* = Sub-analysis

Human Resource Capacity**Quantitative study findings**

Table 3 compared the availability of core clinical staff – doctors, nurses, midwives,

and community health workers in accredited and unaccredited private and government facilities. Fifteen (41.6%) of the accredited private health facilities had two permanent physicians each with a range (min-max) of 1

to 6 while 18 (38.3%) of the unaccredited private facilities had one permanent physician each with a range of 0 to 6. For the government facilities, 3 (23.1%) of the accredited facilities had two permanent physicians each with a range of 0 to 26 while 6 (46.2%) had one temporary physician each. Most of the unaccredited government facilities did not have permanent physicians (10, 76.9%) while all the unaccredited government facilities had no temporary physician. The distributions for other core clinical staff including registered nurses, registered nurse-midwife, community health officers, and community health extension workers are shown in the table.

The complement of permanent and temporary non-core clinical staff - pharmacists, pharmacy technicians,

laboratory scientists, health attendants, and ward orderlies is shown in table 4. Many accredited and unaccredited private and government facilities did not have permanent or temporary non-core clinical staff categories assessed in the study. The situation was worse in the unaccredited facilities; 48.9% to 97.9% of unaccredited private facilities had no single non-core workers whether temporary or permanent, and 69.2% to 100.0% of unaccredited government facilities did not have 75% of the cadres of non-core workers assessed. For accredited facilities, 17 (47.2%) private facilities still had a permanent laboratory technician each and 30.8% to 76.9% of government facilities had one or two staff in about 70% of assessed non-core clinical staff cadres.

Table3: Human resources availability in selected health facilities by ownership and accreditation status: Core clinical staff

	Private Facilities				Government Facilities			
	Accredited (N = 36)		Unaccredited (N = 47)		Accredited (N = 13)		Unaccredited (N = 13)	
	Permanent (n, %)	Temporary (n, %)	Permanent (n, %)	Temporary (n, %)	Permanent (n, %)	Temporary (n, %)	Permanent (n, %)	Temporary (n, %)
Physicians								
Mode	2 (15, 41.6)	0 (11, 30.6)	1 (18, 38.3)	0 (17, 36.2)	2 (3, 23.1)	1 (6, 46.2)	0 (10, 76.9)	0 (13, 100.0)
Range – Min - Max.	1 – 6	0 - 6	0 - 6	0 – 6	0 - 26	0 - 10	0 - 1	-
Registered Nurse								
Mode	2 (10, 27.8)	0 (30, 83.3)	0 (11, 23.4)	0 (38, 80.9)	2 (6, 46.2)	0 (6, 46.2)	2 (5, 38.5)	0 (12, 92.3)
Range – Min - Max.	0 – 9	0 - 4	0 - 7	0 - 4	0 - 106	0 - 5	0 - 3	0 - 1
Registered Nurse-Midwife								
Mode	2 (12, 33.3)	0 (33, 91.7)	0 (26, 55.3)	0 (42, 89.4)	1 (5, 38.5)	0 (11, 84.6)	1 (8, 61.5)	0 (12, 92.3)
Range – Min - Max.	0 – 3	0 - 1	0 - 4	0 - 4	0 - 20	0 - 2	0 - 3	0 - 1
Community Health Officers (CHO)								
Mode	0 (28, 77.8)	0 (36, 100.0)	0 (39, 83.0)	0 (46, 97.9)	0 (13, 100.0)	0 (13, 100.0)	0 (12, 92.3)	0 (13, 100.9)
Range – Min - Max.	0 – 2	-	0 - 3	0 - 1	-	-	0 - 1	-
Community Health Extension Workers (CHEW)								
Mode	0 (14, 38.9)	0 (35, 97.2)	0 (17, 36.2)	0 (44, 93.6)	0 (8, 61.5)	0 (12, 92.3)	0 (10, 76.9)	0 (12, 92.3)
Range – Min - Max.	0 – 14	0 - 1	0 - 6	0 - 6	0 - 4	0 - 1	0 - 1	0 - 1
Auxiliary nurse*								
Mode	0 (14, 38.9)	0 (35, 97.2)	3 (10, 21.3)	0 (43, 91.5)	0 (13, 100.0)	0 (13, 100.0)	0 (12, 92.3)	0 (12, 92.3)
Range – Min - Max.	0 – 16	0 - 2	0 - 10	0 - 2	-	-	0 - 1	0 - 1

(n = modal frequency; % = modal frequency/N)

Table 5 summarizes the total numbers of permanent and temporary staff in the selected facilities. The number of permanent staff of accredited facilities were generally higher than those of unaccredited facilities. For example, 11 (30.6%) of the accredited private facilities compared with 8 (17.0%) of unaccredited private facilities had 20 or more permanent staff, likewise, 8 (61.5%) of accredited government facilities compared with 2 (15.4%) of unaccredited government facilities had 20 or more permanent staff. Contrariwise, there were more temporary staff in the unaccredited facilities than the accredited facilities. The differences in the numerical distribution of permanent staff only and temporary staff only between accredited and unaccredited government facilities was statistically significant (p values of 0.016 and 0.047 respectively),

however, there was no statistically significant difference in such distribution in private facilities.

Qualitative Study Findings

Adequacy of manpower in the health facility: Some of the interviewees opined that there was adequate staff on ground, while others stated that there was inadequate staff:

"I think is a general thing throughout the state, there is the shortage of staffs. ... in this hospital we have just 2 doctors, how would they run steady 24 hours required when they do OPD, 8 am to 4 pm attending to patients, and they come back again for calls? Some of them would go into the theatre to do surgery, how do we cope with 2 doctors and maybe six nurses or 8 nurses apart from the interns and the youth corpsers we sometimes have." (Pharmacist, Public facility).

"Recently they just gave us some staff to augment the ones the state government is having on the ground so that there would be no gap in terms of the manpower that we have in the hospital. They just employed one gardener, a night watchman, and three nurses from OHIS. We are still expecting more from them." (Public facility, Facility Head).

According to a participant in a public health facility, *"...in this department that we have to send our monthly statistics to OHIS and HMB, we need more hands, at least two or three persons should be okay for us, I don't know for other departments I think they will speak for their self."* (Public facility, Record Keeper).

Effects and consequences of manpower shortage in the healthcare facilities: The interviewees stated the consequences of the manpower problems:

"Sometimes we remain here after four o'clock in the evening just to make sure we satisfy some of those patients. So, the workload on us is so much and ...". (Public facility, Pharmacist).

"As I said earlier, we are not working with the people we need now for example at the pharmacy for those dispensing the drugs, we have only one person doing that, which is not supposed to be, so clients have to wait and, in this situation, the client will not be able to leave on time, so, this prolongs the waiting hours." (Public facility, Record Officer).

Impact of OHIS on staffing in the health facilities: Despite the staff challenges in the hospitals across the state, the interviewees unanimously acknowledged the positive impacts of OHIS in their facilities in providing opportunities for engaging additional staff to attend to their clients, leading to increased patient satisfaction:

"...we have thousands of people under OHIS here, so that requires for more staff so, as you can see we have many nurses at their nursing station so that there won't be any delay in attending to these people when they come,so if they come like that in large numbers... so we improved on our staff number so that there won't be a delay for patients when they visit our facility they will be attended to on time, so that is one of the things that we have done" (Private facility, OHIS Desk Officer).

"We used to have the challenges of not having enough staff before we started OHIS, but with the inception of OHIS we had to employ more staff" (Pharmacist, Private facility).

Impact of OHIS on quality of care: The facilities' involvement in the OHIS was reported to have a positive impact on the quality of care. Many claim that they are more able to easily mobilize funds and commodities for their operations:

"The other advantages are that we are equally doing NHIS, the only difference is that they pay us our dues promptly, they pay it monthly (do you mean OHIS or NHIS) I mean OHIS. They pay us every month; we too are taking care of their enrollees very well" (Private facility, Pharmacist).

Influence on OHIS on availability of

consumables and supplies: The scheme has improved availability of consumables in the facilities as attested to by the interviewees:

"It has improved, because for OHIS, some of the consumables that we do buy for OHIS patients, most of the time we do use it for the general patient too. Yeah. For instance, if you buy a commodity now, and the commodity is supposed to expire in a few days, will you allow that commodity to expire rather than use it for the services of the populace, there is a way we use to use it to buffer for other patients and even it's like win-win, some of the donor agencies too,..." (Public Facility, Laboratory Scientist).

Table 4: Human resources availability in selected health facilities by ownership and accreditation status:
Support staff

	Private Facilities				Government Facilities			
	Accredited (N = 36)		Unaccredited (N = 47)		Accredited (N = 13)		Unaccredited (N = 13)	
	Permanent (n, %)	Temporary (n, %)	Permanent (n, %)	Temporary (n, %)	Permanent (n, %)	Temporary (n, %)	Permanent (n, %)	Temporary (n, %)
Pharmacists								
Mode	0 (26, 72.2)	0 (34, 94.4)	0 (40, 85.1)	0 (41, 87.2)	1 (5, 38.5)	0 (6, 46.2)	0 (13, 100.0)	0 (13, 100.0)
Range – Min - Max.	0 - 1	0 - 1	0 - 2	0 - 1	0 - 4	0 - 6	-	-
Pharmacy Technicians								
Mode	0 (18, 50.0)	0 (34, 94.4)	0 (38, 80.9)	0 (43, 91.4)	1 (10, 76.9)	0 (13, 100.0)	1 (11, 84.6)	0 (13, 100.0)
Range – Min - Max.	0 - 4	0 - 1	0 - 2	0 - 2	0 - 1	-	0 - 1	-
Laboratory Scientists								
Mode	0 (18, 50.0)	0 (28, 77.)	0 (30, 63.8)	0 (41, 87.2)	1 (6, 46.2)	0 (9, 69.2)	0 (10, 76.9)	0 (13, 100.0)
Range – Min - Max.	0 - 2	0 - 3	0 - 2	0 - 2	0 - 30	0 - 10	0 - 1	-
Laboratory Technicians								
Mode	1 (17, 47.2)	0 (35, 97.2)	0 (35, 74.5)	0 (43, 91.5)	0 (9, 69.2)	0 (10, 76.9)	0 (12, 92.3)	0 (13, 100.0)
Range – Min - Max.	0 - 6	0 - 1	0 - 3	0 - 1	0 - 24	0 - 4	0 - 1	0
Record Officers								
Mode	0 (16, 44.4)	0 (36, 100.0)	0 (31, 66.0)	0 (44, 93.6)	1 (4, 30.8)	0 (5, 38.9)	1 (9, 69.2)	0 (12, 92.3)
Range – Min - Max.	0 - 6	-	0 - 2	0 - 1	0 - 1	0 - 5	0 - 2	0 - 2
Health Attendants								
Mode	0 (12, 33.3)	0 (34, 94.4)	0 (23, 48.9)	0 (46, 97.9)	2 (4, 30.8)	0 (11, 84.6)	2 (7, 53.8)	0 (13, 100.0)
Range – Min - Max.	0 - 38	0 - 8	0 - 10	0 - 1	0 - 4	0 - 8	1 - 5	-
Ward Orderlies								
Mode	0 (21, 58.3)	0 (35, 97.2)	0 (26, 55.3)	0 (46, 97.9)	0 (10, 76.9)	0 (12, 92.3)	0 (9, 69.2)	0 (13, 100.0)
Range – Min - Max.	0 - 11	0 - 2	0 - 13	0 - 1	0 - 7	0 - 1	0 - 2	-

(N = modal frequency; % = modal frequency/N)

DISCUSSION

This study assessed the level of medical manpower in facilities enrolled by the Osun Health Insurance Scheme (OHIS), and compared it with the available manpower in the OHIS non-accredited facilities.

Majority of healthcare facilities in this study were hospitals with less than 20 beds. The hospitals in the state, and the country at large, are mostly small sized with

few bed capacities run mostly by private individuals (sole proprietorship).. This problem has been well documented in global statistics with Nigeria battling with a bed density of as low as less than 10 per 10000 population, the figure which is far lower than the global average of 27 per 10000 population.⁹

Our study clearly highlighted a gross shortage of healthcare personels across facilities. Generally, in this study, accredited

health facilities, private or government, had more core clinical staff compared to unaccredited facilities but it is important to note that some accredited facilities had no registered nurse or nurse-midwife. Health insurance provides means to achieving universal health coverage (UHC) for full access to quality healthcare.⁴ However, inadequacies of clinical staffing has been identified as a major barrier to effective implementation of health insurance, a study identified inadequacy of health personnel as a key problem militating against implementation of the national health insurance across hospitals in the capital city of Nigeria.¹⁰

This study also noted more permanent workers in the accredited facilities, and more temporary workers in the unaccredited facilities; this is not unexpected because OHIS enrolment criteria include availability of key permanent hospital staff. It is a common knowledge that hospitals in Nigeria are increasingly engaging health workers on locum basis as a stopgap response to the shortage being fueled by mass exodus of healthcare workers. Temporary staff are often not well-motivated and do not always show desired commitment to duty, and this can have grave consequences on the continuity and quality of care.¹¹ The widespread shortage of healthcare professionals is not a local problem but an issue that cut across the country. Nigeria as a whole has had to contend with the burden of severe shortage of skilled healthcare workers; the country has a density of 1.83 skilled healthcare workers per thousand people which is way short of the World Health Organisation (WHO) recommended 4.45 per thousand people.¹² Using physicians to emphasize the enormity of this problem for example, there are currently 3.8 doctors per 10,000 population in Nigeria, and this is less than a quarter of the WHO recommended 16.6 doctors per 10,000 population.¹³ With a population far above 235 million, and growing exodus of medical professionals from the country, this problem will become a national crisis if not effectively

addressed.^{7,14} As a step in the right direction to mitigate the looming manpower crisis in Nigeria, the Nigerian government recently came up with a national policy on health workforce migration with an overarching objective of curbing the brain drain by providing the conditions and environment which will encourage the workers to stay in the country and inspire the return of healthcare professionals from abroad.¹⁵

This study reported positive impact of OHIS; there were more funding and staffing in the health facilities enrolled in the scheme. This findings aligns with the general knowledge that health insurance schemes generally improve hospitals' income which can be invested into staffing and capacity development of staff, in addition to improving expenditures in other core areas. A study incorporating 15 secondary health facilities in northern Nigeria reported up to three-fold increase in income of hospitals following enrolment into health insurance scheme with the fund from the scheme adding up to a third of the total revenue of the health facilities, leading to improvement in staff strength, infrastructure and capacities.¹ This study highlighted that despite the manpower challenges in the hospitals across the state, there were unanimously acknowledged positive impacts of OHIS in areas of staffing, hospital supplies and overall quality of care.

The study concludes that there was widespread shortage of healthcare manpower, the situation was better in OHIS accredited facilities. This study established that OHIS provides extra funds for healthcare facilities translating to better staffing and more supplies and consumables. This study recommends concerted efforts for improved staffing of healthcare facilities in Osun State, and for OHIS to spread its network to incorporate many more facilities for efficient healthcare delivery and enhanced quality of care.

Conflict of interest: Authors declare that they have no conflict of interest.

Table 5: Total number of permanent and temporary staff complement in selected health facilities by ownership and accreditation status

	Private Facilities			Government Facilities		
Number of permanent staff only	Accredited (n= 36) n (%)	Unaccredited (n= 47) n (%)	Statistics	Accredited(n= 13) n (%)	Unaccredited (n= 13) n (%)	Statistics
<10	9 (25.0)	22 (46.8)	$\chi^2=1.394$	5 (38.5)	1 (7.7)	$\chi^2=5.754$
10 – 19	16 (44.4)	17 (36.2)	df=2	1 (7.7)	10 (76.9)	df=2
20+	11 (30.6)	8 (17.0)	p-value=0.238	8 (61.5)	2 (15.4)	p value=0.016*
Total number of temporary staff only						
<10	35 (97.2)	45 (95.7)	$\chi^2=0.137$	8 (61.5)	2 (15.5)	$\chi^2=3.926$
10 – 19	0 (0.0)	1 (2.1)	df=2	3 (23.1)	1 (7.7)	df=2
20+	1 (2.8)	1 (2.1)	p-value=0.718	2 (15.4)	10 (76.9)	p value=0.047*

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